

Notes: Howell, Kuchler, Snitkof, Stroebe, and Wong (JF, 2024)

Lender Automation and Racial Disparities in Credit Access

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Why the PPP is a powerful setting
- 2.2 PPP terms and sample construction
- 2.3 Lender classification and automation measures
- 2.4 Measuring borrower race and ethnicity
- 2.5 Core outcomes and explanatory variables
- 2.6 Racial-animus measures
- 2.7 Descriptive facts that frame the paper

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Mechanisms and testable implications
- 3.2 Cross-lender linear-probability specification
- 3.3 Dynamic event-study specification for bank automation
- 3.4 Post-automation diff-in-diff specification
- 3.5 Specifications for racial animus
- 3.6 Application-level regressions using Lendio
- 3.7 Relationship-banking regressions using Ocrolus
- 3.8 Estimation choices and standard errors

4 Identification and instruments (what makes the estimates credible)

- 4.1 Why the PPP is unusually clean for this question
- 4.2 What identifies the cross-lender estimates
- 4.3 Why the within-bank automation design is the cleanest causal evidence
- 4.4 Loan-size and geography channels are real but incomplete
- 4.5 Evidence for the discrimination channel
- 4.6 Application behavior versus approval behavior
- 4.7 Checking-account and credit relationships do not explain the gap
- 4.8 Performance and fraud are not persuasive alternatives
- 4.9 What the paper can and cannot distinguish
- 4.10 External-validity limits

5 Tables and figures

5.1 Table I and Table II: sample composition and why small-loan economics matter . . .

5.2 Figure 1: unconditional lender shares for Black-owned firms

5.3 Table III and Figure 2: bank automation event study

5.4 Table IV and Figure 3: how much controls explain, and where the remaining gap sits

5.5 Table V and Table VI: automation, racial animus, and the discrimination channel . .

5.6 Table VII: Lendio application data

5.7 Table VIII and Table IX: relationship banking does not rescue the bank story

6 Short summary

7 Conclusion

7.1 Core conclusion (what the paper establishes)

7.2 Economic intuition

7.3 Takeaway

1 Big picture

- **Question.** Does lender automation reduce racial disparities in credit access, and if so, through what channels?
- **Setting.** The paper studies Paycheck Protection Program (PPP) lending during COVID. This is unusually useful because PPP loans had fixed terms and were 100% government guaranteed, so lenders bore essentially no credit risk.
- **Core challenge.** Black-owned firms borrowed disproportionately from fintech lenders and disproportionately less from small conventional banks. But this pattern could reflect many things besides discrimination or automation: smaller loan size, branch-network geography, different application behavior, relationship banking, real-time firm performance, or fraud.
- **Main conceptual contribution.** The paper separates *process automation* from algorithmic underwriting. The key idea is that automating intake, payroll verification, compliance checks, and application routing can change who gets served even when the lender is not pricing credit risk.
- **Main mechanisms.** Automation can raise lending to Black-owned firms through three channels:
 - lower fixed cost per loan, which makes small loans profitable;
 - broader geographic reach through online origination, which matters in underbanked minority areas;
 - less human discretion in triage and processing, which reduces room for taste-based discrimination.
- **Main empirical designs.** The paper uses two complementary designs:
 - *cross-lender comparisons* across fintechs, top-4 banks, other large banks, and small/medium banks;
 - *within-bank event-study / diff-in-diff evidence* from small and medium banks that automated during PPP.
- **Main unconditional fact.** Fintech lenders made 26.5% of their PPP loans to Black-owned firms, versus only 3.3% for small banks and 6.2% for top-4 banks.
- **Main market-share fact.** Fintechs originated 53.6% of all PPP loans to Black-owned firms while accounting for only 17.4% of all PPP loans.
- **Main controlled cross-lender result.** After controlling for approval week, loan size, ZIP code, industry, business type, and employer status, Black-owned firms remain 12.1 percentage points more likely to get a fintech PPP loan.
- **Main causal within-bank result.** When small and medium banks automate during PPP, their Black-owned loan share rises by 6 percentage points in the simple diff-in-diff and by 4.3 percentage points after rich controls, relative to a preautomation Black share of 4.4%.
- **Main discrimination result.** The residual Black-owned shift toward fintechs is larger in places with more anti-Black racial animus, and the positive within-bank effect of automation is also larger in high-animus areas.

- **Main application-result.** In Lendio application data, Black-owned firms routed only to fin-techs show no approval gap, but Black-owned firms routed only to conventional lenders are 3.9 percentage points less likely to get a PPP loan from that lender and 5.8 percentage points more likely to get no PPP loan at all.
- **Main relationship-banking result.** Preexisting checking-account and credit relationships do not explain the cross-lender racial disparity. The disparity is concentrated among firms that needed to establish a *new* relationship for PPP borrowing.
- **Economic takeaway.** Automation appears to reduce racial disparities in small-business credit access by making small loans economical, expanding reach beyond branch networks, and limiting the effect of human bias in processing decisions.

2 Data and measurement (key objects)

2.1 Why the PPP is a powerful setting

- PPP loan size was mechanically tied to payroll: a borrower could receive 2.5 times monthly pre-COVID payroll.
- Contract terms were fixed rather than negotiated, so there is no variation in interest rates, collateral, or maturities that lenders could use to screen borrowers.
- PPP loans were fully guaranteed by the federal government, so lenders bore essentially no default risk.
- Lenders were still free to decide *which* completed applications to process when applications vastly exceeded capacity.

Interpretation. This removes the usual confound that racial lending disparities may reflect expected default or contract pricing. In PPP, the central lender decision is closer to a *service / processing* decision than a credit-risk decision.

2.2 PPP terms and sample construction

- Raw SBA data contain 11.8 million PPP loans through May 31, 2021.
- Dropping second-draw loans leaves 8.8 million first-draw loans.
- The baseline analysis then drops loans after February 23, 2021, when program rules changed sharply to prioritize very small firms and certain mission-oriented lenders.
- The resulting baseline sample contains 5.7 million first-draw PPP loans.

Other institutional details matter for interpretation:

- Forgiveness turned the loan into a grant if payroll and other eligible-spending conditions were satisfied.
- Lenders were paid upfront fees equal to 5% for loans up to \$350,000, 3% for loans between \$350,000 and \$2 million, and 1% above \$2 million.

- This fee schedule made smaller PPP loans less attractive to lenders with higher processing costs.

Interpretation. The PPP fee schedule gives a concrete reason why low-cost automated lenders might be disproportionately willing to serve very small firms.

2.3 Lender classification and automation measures

The paper classifies PPP originators into eight mutually exclusive groups:

- top 4 banks,
- other large banks,
- medium banks,
- small banks,
- credit unions,
- CDFIs/nonprofits,
- minority depository institutions,
- fintech lenders.

Key data facts from Table I, Panel B:

- banks still originated 75% of PPP loans in the analysis sample;
- fintech lenders originated 17.4% of loans overall;
- mean fintech loan size was only \$31,228 versus \$88,083 at small banks and \$146,505 at large banks.

Automation is measured in two complementary ways:

- **across lenders:** lender type itself proxies for the degree of process automation, with fintechs at the top and small banks at the bottom;
- **within conventional banks:** branch-level software spending from Spiceworks Ziff Davis for 2017–2019, covering 108,099 branches of 3,440 lenders.

The paper also gathers explicit automation-event dates for 20 small and medium banks that automated during the PPP, largely through Biz2Credit or similar back-end providers.

Interpretation. The paper is strongest because automation is not measured only by a broad lender-type label. It also uses a within-bank software-spending proxy and a sharp before/after automation event for a subset of banks.

2.4 Measuring borrower race and ethnicity

A central step is assigning race to firm owners.

- Owner names come from Middesk secretary-of-state filings, business names for many nonemployers, and Lendio applications.

- Step 1 uses surname and census-tract location to form a Bayesian posterior race distribution.
- Step 2 feeds that posterior, plus first name, industry, and employer status, into a random forest with 1,000 trees.

The final measure predicts whether the owner is Asian, Black, Hispanic, or White.

Important facts:

- the paper predicts race for 4.18 million PPP borrowers;
- in a holdout sample, 84% of owners predicted to be Black self-identify as Black.

Interpretation. Predicted race is arguably the relevant object here because manual loan processors observe names and locations, not necessarily self-identified race.

2.5 Core outcomes and explanatory variables

Cross-lender outcomes. The paper studies indicators for whether a borrower's PPP lender is:

- a fintech,
- a top-4 bank,
- another large bank,
- or a small/medium bank.

Within-bank automation outcome. The event-study outcome is an indicator for whether a PPP loan made by bank b in period t goes to a Black-owned firm.

Key explanatory variable. An indicator for whether the borrower is Black-owned.

Main controls.

- approval week fixed effects,
- 100 loan-amount percentile fixed effects,
- ZIP code fixed effects,
- three-digit NAICS industry fixed effects,
- business-type fixed effects,
- employer-status fixed effects.

Additional mechanism data.

- **Lendio:** 278,404 PPP applications quasi-randomly routed to lender sets;
- **Ocrolus:** about 168,000 matched PPP borrowers with bank-statement data, checking-account banks, credit relationships, and cash-flow measures;
- **Enigma:** card-revenue data for more than 800,000 borrowers to proxy for real-time firm performance.

2.6 Racial-animus measures

The discrimination channel is tested with six local measures of anti-Black animus:

- implicit-bias IAT score,
- explicit-bias IAT score,
- racially charged Google searches (Stephens-Davidowitz),
- Nationscape favorability toward Black people,
- residential dissimilarity,
- residential isolation.

Each measure is standardized at its own geography level, so the interaction coefficients can be read as the effect of a one-standard-deviation increase in animus.

Interpretation. This is not just one noisy proxy. The paper shows the same sign pattern across several independent animus measures.

2.7 Descriptive facts that frame the paper

From Tables I and II:

- In the analysis sample, 8.6% of borrowers are predicted Black-owned and 75.0% are White-owned.
- Mean PPP loan size is only \$24,315 for Black-owned firms versus \$110,317 for White-owned firms.
- Among Black-owned firms, 53.6% of loans come from fintech lenders, compared with only 11.4% among White-owned firms.
- Among White-owned firms, 30.2% of loans come from small banks; among Black-owned firms, only 9.8% do.
- Only 21.0% of Black-owned firms are employer businesses, versus 67.6% of White-owned firms.

Interpretation. Black-owned PPP borrowers are smaller on average and far more concentrated among fintech lenders. Any convincing explanation must therefore grapple with both the small-loan margin and the lender-type margin.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mechanisms and testable implications

The paper organizes its economics around three mechanisms:

- **Fixed-cost / capacity channel.** Automation lowers cost per application and expands throughput, so low-fee small loans become worth processing.
- **Geographic-reach channel.** Online origination weakens branch-network dependence, helping firms in underbanked minority areas.
- **Discrimination channel.** Removing human discretion in intake, verification, and prioritization reduces scope for taste-based bias.

There is also a possible offsetting force: manual lenders might sometimes provide more personalized help in completing applications. So the net effect of automation is an empirical question rather than an identity.

3.2 Cross-lender linear-probability specification

The central cross-lender regression is:

$$1(LenderType_i) = \beta 1(BlackOwned_i) + X_i\delta + \varepsilon_i.$$

Here:

- $LenderType_i$ is an indicator for whether borrower i gets its PPP loan from a fintech, top-4 bank, large bank, or small/medium bank;
- $1(BlackOwned_i)$ is the key indicator;
- X_i is the rich vector of loan, firm, time, geography, and industry controls.

Interpretation. The coefficient β asks whether otherwise similar Black-owned firms are more likely to match with a more automated lender.

3.3 Dynamic event-study specification for bank automation

For banks that automate during PPP, the paper estimates:

$$1(BlackOwned_{ibt}) = \sum_{k \neq -1} \beta_k 1(t - A_b = k) + \alpha_b + \alpha_t + \varepsilon_{ibt},$$

where A_b is bank b 's automation date.

- α_b absorbs time-invariant bank differences.
- α_t absorbs common time shocks.
- The omitted period is the month just before automation ($k = -1$).

Interpretation. This is the cleanest causal design in the paper. If automation matters, Black-owned loan shares should jump after the bank automates, with no pretrend before.

3.4 Post-automation diff-in-diff specification

The paper also estimates the simpler controlled diff-in-diff:

$$1(BlackOwned_{ibt}) = \alpha_b + \alpha_t + \beta 1(PostAuto_{bt}) + X_i\delta + \varepsilon_{ibt}.$$

Interpretation. The coefficient β summarizes the average post-automation increase in Black-owned lending share once bank fixed effects, time fixed effects, and borrower controls are included.

3.5 Specifications for racial animus

To test the discrimination channel, the paper interacts Black ownership with local racial animus in the cross-lender regressions, and it interacts post-automation with racial animus in the within-bank diff-in-diff.

Schematic form:

$$1(Fintech_i) = \beta_1 1(BlackOwned_i) + \beta_2 (1(BlackOwned_i) \times Animus_g) + X_i\delta + FE + \varepsilon_i.$$

Interpretation. If the remaining lender gap reflects taste-based bias at less automated lenders, then Black-owned firms should shift more strongly toward fintechs in high-animus places, and automation should matter more there too.

3.6 Application-level regressions using Lendio

The Lendio design studies completed applications that were quasi-randomly routed to lender sets conditional on loan size, geography, and lender capacity.

The two main outcomes are:

- whether the borrower receives a PPP loan from one of the lenders to which Lendio sent the application;
- whether the borrower receives no PPP loan at all.

The regressions include fixed effects for the exact combination of lenders to which the application was routed.

Interpretation. This is powerful because it sharply reduces the concern that Black-owned firms simply *chose* fintechs more often.

3.7 Relationship-banking regressions using Ocrolus

Using bank-statement data, the paper augments the baseline lender-type regressions with:

- checking-account-bank fixed effects,
- indicators for prior credit relationships with fintech and conventional lenders,
- monthly cash-inflow and net-cash-inflow controls.

Interpretation. These specifications test whether the racial lender gap is really just a relationship-banking story or a firm-performance story.

3.8 Estimation choices and standard errors

- Most specifications are linear-probability models.
- Standard errors are clustered by borrower ZIP code.
- The event study is estimated at the month level; a weekly version is used as a robustness check around automation dates.

Interpretation. The paper keeps the econometrics simple and transparent and does the heavy lifting with institutional design, fixed effects, and multiple complementary data sets.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the PPP is unusually clean for this question

The core identification strength is the PPP environment itself:

- no lender credit exposure,
- standardized loan terms,
- no endogenous pricing,
- huge processing bottlenecks.

Interpretation. If a Black-owned firm is less likely to be served in this setting, the explanation is much less likely to be standard risk-based lending.

4.2 What identifies the cross-lender estimates

The cross-lender regressions compare otherwise similar borrowers within the same approval week, location, industry, loan-size bucket, business form, and employer status.

This matters because it directly controls for the paper's first two mechanisms:

- smaller loans,
- underserved geography.

After these controls, the remaining fintech tilt for Black-owned firms is interpreted as the part not explained by these observable channels.

Interpretation. The residual 12.1 percentage-point fintech effect is not the whole story of automation, but it is the part most suggestive of something beyond size and location.

4.3 Why the within-bank automation design is the cleanest causal evidence

The event-study design compares the same bank before and after automation against otherwise similar nonautomating banks.

Credibility comes from three facts:

- no differential pretrends before automation;
- a discrete break after automation;
- the bank-facing website did not change materially, so the treatment is mainly back-end processing automation.

Interpretation. This design is much less vulnerable than the cross-lender design to concerns about persistent lender culture or customer selection.

4.4 Loan-size and geography channels are real but incomplete

The paper shows that automation works partly through real economic channels:

- controlling for loan size reduces the Black-owned fintech coefficient from 26.0 to 23.2 percentage points;
- controlling for ZIP code reduces it from 26.0 to 20.7 percentage points;
- adding industry, business type, and employer status reduces it further to 12.1 percentage points.

Interpretation. Small-loan economics and branch-network geography explain a lot, but not all, of the racial lender gap.

4.5 Evidence for the discrimination channel

The PPP setting makes *statistical discrimination* much less plausible because lenders did not bear default risk. That pushes interpretation toward *taste-based discrimination* if a residual racial gap remains.

The paper's discrimination evidence is layered:

- residual Black-owned substitution toward fintechs is larger in high-animus areas;
- Black-owned firms are especially unlikely to get non-top-4 bank loans in high-animus areas;
- automation has a larger within-bank effect in high-animus places;
- top-4 banks show much less evidence of an animus-sensitive Black-owned lending gap.

Interpretation. The animus interactions do not prove discrimination on their own, but in a no-credit-risk environment they are very hard to rationalize with purely mechanical lending technology stories.

4.6 Application behavior versus approval behavior

One obvious alternative explanation is that Black-owned firms simply applied to fintechs more often. The Lendio evidence sharply narrows this story:

- when routed only to fintechs, Black-owned firms have essentially no approval disadvantage;
- when routed only to conventional lenders, they are 3.9 percentage points less likely to get the loan from that lender and 5.8 percentage points more likely to get no PPP loan at all;
- when routed only to small banks, the disparities are even larger.

Interpretation. Application differences may contribute in the raw data, but lender-side decisions clearly matter.

4.7 Checking-account and credit relationships do not explain the gap

Banks did favor their own checking-account customers, especially the top 4. But the racial disparity survives even after comparing firms with the *same* checking-account bank.

Key facts:

- In the Oculous sample, Black ownership predicts a 5.5 percentage-point higher probability of a fintech PPP loan, even after rich controls.
- Adding checking-account-bank fixed effects changes that coefficient essentially not at all.
- The main disparity sits among the 72.6% of firms that borrowed from a lender other than their checking-account bank.

Interpretation. The story is not “Black-owned firms banked with the wrong banks.” It is much more about who could successfully form a new relationship under extreme processing pressure.

4.8 Performance and fraud are not persuasive alternatives

The paper explicitly checks two final alternatives:

- **Real-time performance.** Matching more than 800,000 borrowers to Enigma card-revenue data, the paper finds that controlling for current revenue does not attenuate the racial lender gap.
- **Fraud.** The paper finds no evidence that differential fraud rates among Black-owned firms, combined with different compliance intensity across lenders, explain the cross-lender results.

Interpretation. These checks matter because they target the most natural “hard-information” objections that remain once credit risk is taken off the table.

4.9 What the paper can and cannot distinguish

The paper can credibly say:

- automation causally increases Black-owned lending shares within automating banks;
- cross-lender racial patterns line up strongly with automation;
- loan-size, geography, and discrimination-related channels all matter.

But the paper cannot fully pin down the exact decomposition of the residual cross-lender gap across:

- differential lender approval behavior,
- anticipatory self-selection by applicants,
- historical branch-network patterns,
- or other unobserved lender characteristics correlated with automation.

Interpretation. The within-bank automation design is the causal anchor; the other data sets show why the cross-lender pattern looks the way it does.

4.10 External-validity limits

The authors are careful about generalization:

- PPP capacity constraints may have made discrimination especially cheap because rejected Black-owned applications could often be replaced with equally profitable White-owned applications;
- but small banks also moved much more activity online during COVID, so the measured automation gap may understate what would happen in a normal, more face-to-face market.

Interpretation. Outside PPP, the magnitude could be smaller because competition disciplines bias more tightly, or larger because in-person discretion returns.

5 Tables and figures

5.1 Table I and Table II: sample composition and why small-loan economics matter

Table I

Summary Statistics by Lender Type

This table reports summary statistics for PPP loans by originating lender type. Panel A describes all PPP loans, while Panel B focuses on one statistical sample, which is composed of first draw PPP loans between April 1, 2020, and February 28, 2021, for which we can predict the borrower's race. All subsequent statistics and analyses are drawn from subsamples of the data defined in Panel B. Panel C reports statistics for banking and credit relationships as well as financial performance for borrowers in the analysis sample that we can match to bank statement data from Dun & Bradstreet. The columns "Share Checking Account with PPP Lender" capture what share the borrower's business banking account bank is the same institution that originated the PPP loan. The remaining variables are derived from transactions at the borrower's most recent monthly bank statement. Internal Appendix Table 1A.11 reports Panels B and C for the subset without predicted race.

Panel A: All PPP loans									
PPP Loan Amount (\$)									
	Number Lenders	Number Loans	Total Amt. \$ Bil.	Mean	P10	P50	P90		
All	5,203	11,706,420	796.7	67,867	4,399	39,679	151,656		
Top 4	4	1,270,903	94.6	75,395	5,510	31,378	148,241		
Large Banks	10	696,115	198.6	121,127	6,000	36,133	201,695		
Medium Banks	430	1,532,696	211.4	126,977	4,006	36,940	206,140		
Small Banks	1,668	2,020,637	191.9	75,990	4,165	39,402	151,646		
Credit Unions	569	170,673	15.7	41,479	3,063	15,982	62,452		
CDFI/Nonprofit	201	1,010,816	14.4	22,111	4,400	20,040	91,603		
Minority Dep Inst	113	186,072	16.5	17,483	3,213	17,100	103,490		
Fintech	10	1,796,012	65.6	65,738	3,063	17,209	95,168		(Continued)

Table I—Continued

Panel B: Analysis sample

This table reports summary statistics for PPP loans by originating lender type. Panel A describes all PPP loans, while Panel B focuses on one statistical sample, which is composed of first draw PPP loans between April 1, 2020, and February 28, 2021, for which we can predict the borrower's race. All subsequent statistics and analyses are drawn from subsamples of the data defined in Panel B. Panel C reports statistics for banking and credit relationships as well as financial performance for borrowers in the analysis sample that we can match to bank statement data from Dun & Bradstreet. The columns "Share Checking Account with PPP Lender" capture what share the borrower's business banking account bank is the same institution that originated the PPP loan. The remaining variables are derived from transactions at the borrower's most recent monthly bank statement. Internal Appendix Table 1A.11 reports Panels B and C for the subset without predicted race.

Panel B: Analysis sample									
PPP Loan Amount (\$)									
	Number Lenders	Number Loans	Total Amt. \$ Bil.	Mean	P10	P50	P90	Share Bank Share	Share White Share
All	5,206	4,358,023	361.9	69,666	4,402	20,851	174,906	6.4%	15.0%
Top 4	4	881,471	52.0	79,911	6,001	21,741	141,487	6.2%	61.0%
Large Banks	10	589,084	171.7	146,050	6,000	32,199	279,287	5.2%	61.0%
Medium Banks	420	1,517,761	146.5	144,654	4,001	30,691	202,266	4.1%	64.0%
Small Banks	1,608	1,984,368	80.7	69,093	4,002	20,851	173,948	5.3%	69.0%
Credit Unions	569	173,368	1.9	44,612	3,000	16,891	69,427	3.1%	78.1%
CDFI/Nonprofit	200	81,576	5.9	71,264	3,001	20,851	105,003	19.6%	74.6%
Minority Dep Inst	114	81,663	5.0	64,001	4,001	20,851	100,489	4.0%	67.0%
Fintech	10	799,084	22.7	71,229	5,003	15,916	94,685	39.6%	67.0%

(Continued)

Table I—Continued

Panel C: Bank and credit relationships sample (Ocrulus)

This table reports summary statistics for PPP loans by originating lender type. Panel A describes all PPP loans, while Panel B focuses on one statistical sample, which is composed of first draw PPP loans between April 1, 2020, and February 28, 2021, for which we can predict the borrower's race. All subsequent statistics and analyses are drawn from subsamples of the data defined in Panel B. Panel C reports statistics for banking and credit relationships as well as financial performance for borrowers in the analysis sample that we can match to bank statement data from Dun & Bradstreet. The columns "Share Checking Account with PPP Lender" capture what share the borrower's business banking account bank is the same institution that originated the PPP loan. The remaining variables are derived from transactions at the borrower's most recent monthly bank statement. Internal Appendix Table 1A.11 reports Panels B and C for the subset without predicted race.

Panel C: Bank and credit relationships sample (Ocrulus)									
RSE Has									
	Number Lenders	Number Loans	Total Amt. \$ Bil.	Mean	P10	P50	P90	Share Bank Share	Share White Share
All	109,360	80,897	27.4%	14.2%	80.0%	9,016	1,302	15.3%	63.0%
Top 4	29,700	72,700	47.2%	14.8%	81.0%	11,000	1,770	14.4%	63.0%
Large Banks	15,170	104,000	48.6%	14.1%	81.0%	12,000	2,400	13.1%	74.0%
Medium Banks	26,430	120,770	38.4%	14.1%	79.0%	9,711	1,400	13.0%	79.0%
Small Banks	22,010	104,000	23.3%	14.1%	79.0%	7,700	1,200	13.1%	79.0%
Credit Unions	1,000	86,300	14.3%	13.6%	79.0%	4,700	800	13.6%	80.0%
CDFI/Nonprofit	1,000	86,300	14.3%	13.6%	79.0%	4,700	800	13.6%	80.0%
Minority Dep Inst	1,000	86,300	14.3%	13.6%	79.0%	4,700	800	13.6%	80.0%
Fintech	1,000	86,300	14.3%	13.6%	79.0%	4,700	800	13.6%	80.0%

(Continued)

Read:

- Table I shows that fintech lenders originated 17.4% of all PPP loans in the analysis sample, but their average loan size was only \$31,228.
- Small banks originated 25.4% of all loans, with an average loan size of \$88,083.
- Table II shows how stark the racial composition is: 53.6% of Black-owned PPP loans came from fintechs, compared with only 11.4% for White-owned firms.
- Black-owned firms also had far smaller average PPP loans: \$24,315 versus \$110,317 for White-owned firms.

Table II

Summary Statistics by Predicted Race

This table reports summary statistics by race/ethnicity. Panel A contains loan and firm characteristics for the analysis sample. Panel B summarizes information from the bank statement-matched data.

Panel A: Analysis sample					
	All	Asian-Owned	Black-Owned	Hispanic-Owned	White-Owned
Loan Amount					
Mean Loan Amount (\$)	93,666	53,492	24,315	54,287	110,317
Median Loan Amount (\$)	20,833	20,071	14,886	16,218	23,700
Share Total Loans Made by Bank Types					
Top 4	15.9%	27.4%	11.4%	24.1%	14.2%
Large Banks	9.4%	8.1%	5.8%	6.9%	10.2%
Medium Banks	24.3%	18.1%	11.5%	17.1%	27.3%
Small Banks	25.4%	12.4%	9.8%	11.7%	30.2%
Credit Unions	4.1%	2.4%	4.7%	3.8%	4.3%
CDFI/Nonprofit	2.0%	1.7%	2.4%	1.9%	2.0%
Minority Dep Inst	1.5%	8.0%	0.7%	4.0%	0.5%
Fintech	17.4%	21.7%	53.6%	30.5%	11.4%
Business Types					
Corporation	27.8%	40.3%	11.2%	28.7%	28.1%
LLC	26.8%	24.4%	19.9%	23.1%	28.1%
Other	11.9%	8.3%	26.2%	15.8%	10.3%
Sole Proprietorship	20.4%	14.9%	38.4%	21.5%	18.8%
Subchapter S Corporation	13.1%	12.0%	4.3%	10.1%	14.6%
Employer Institution	63.4%	73.8%	21.0%	57.8%	67.6%
Industries (3 Digit NAICS)					
Professional/Technical Services	12.7%	7.8%	10.6%	10.8%	13.7%
Ambulatory Health Care Services	7.5%	10.3%	8.2%	6.3%	7.2%
Food and Drinking Services	5.9%	14.7%	3.5%	8.7%	4.8%
Personal and Laundry Services	6.0%	12.0%	15.6%	7.5%	4.0%
Specialty Trade Contractors	5.4%	0.9%	2.2%	6.3%	6.2%
Other	62.6%	54.3%	60.0%	60.4%	64.1%
Observations	4,183,623	372,993	359,366	313,389	3,137,875

(Continued)

Table II—Continued

Panel B: Bank and credit relationships sample (Ocrulus)

This table reports summary statistics for PPP loans by originating lender type. Panel A describes all PPP loans, while Panel B focuses on one statistical sample, which is composed of first draw PPP loans between April 1, 2020, and February 28, 2021, for which we can predict the borrower's race. All subsequent statistics and analyses are drawn from subsamples of the data defined in Panel B. Panel C reports statistics for banking and credit relationships as well as financial performance for borrowers in the analysis sample that we can match to bank statement data from Dun & Bradstreet. The columns "Share Checking Account with PPP Lender" capture what share the borrower's business banking account bank is the same institution that originated the PPP loan. The remaining variables are derived from transactions at the borrower's most recent monthly bank statement. Internal Appendix Table 1A.11 reports Panels B and C for the subset without predicted race.

Panel B: Bank and credit relationships sample (Ocrulus)					
	All	Asian-Owned	Black Owned	Hispanic-Owned	White-Owned
Banking Relationships					
Share Checking Account from Top 4 Banks	47.8%	57.5%	47.1%	61.9%	44.1%
Share Checking Account from Large Banks	15.8%	13.6%	19.1%	11.6%	16.0%
Share Checking Account from Small/Medium-Sized Banks	27.5%	20.3%	16.5%	19.9%	32.7%
Share Checking Account from Others	8.9%	8.7%	17.2%	6.7%	7.2%
Credit Relationships					
Share Credit Relationship with Fintech	14.2%	13.2%	7.9%	12.5%	16.2%
Share Credit Relationship with Non-Fintech	80.0%	78.9%	81.4%	85.8%	78.9%
Share Credit with checking account bank	17.6%	18.4%	14.4%	19.5%	17.9%
Cash In/Outflows					
Mean Monthly Cash Inflow (\$)	231,390	223,615	81,706	201,382	273,948
Median Monthly Cash Inflow (\$)	65,232	70,141	11,636	58,644	87,949
Mean Monthly Cash Outflow (\$)	218,738	213,055	72,270	189,775	260,004
Median Monthly Cash Outflow (\$)	60,890	67,053	8,960	52,466	82,450
Mean Monthly Net Cash Inflow (\$)	9,016	7,426	6,332	8,681	9,977
Mean Monthly Net Cash Inflow (\$)	1,332	1,232	703	1,698	1,976
Observations	168,360	17,166	25,782	18,511	106,881

Economic message. The raw facts already point to the small-loan channel. Black-owned firms are smaller, and automated lenders are exactly the lenders best positioned to process small PPP loans profitably.

5.2 Figure 1: unconditional lender shares for Black-owned firms

Read:

- Panel A shows the share of each lender type's own PPP loans that went to Black-owned firms.
- Panel B shows which lender types accounted for total Black-owned PPP lending.

- Fintechs dominate both views of the data: they lend much more often to Black-owned firms and account for more than half of all Black-owned PPP loans.

Economic message. Figure 1 is the paper’s motivating fact. Any explanation of PPP racial disparities must explain why fintechs and small banks look so different.

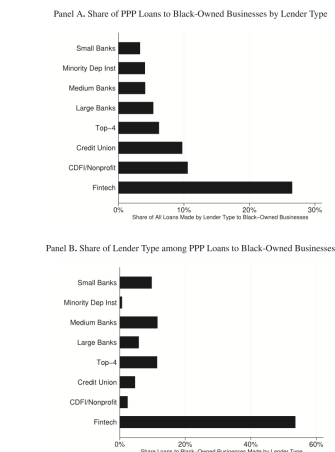


Figure 1. Black-owned business PPP lending by lender type. Panel A shows, for each lender type, the share of PPP loans that went to Black-owned businesses (P(Black-owned|Originating Lender Type)). Panel B shows the share of all PPP loans to Black-owned businesses made by each lender type (P(Originating Lender Type|Black-owned)).

Table III
Summary Statistics on Automation during PPP among Small and Medium-Sized Banks

This table contains summary statistics for the banks used in the automation analysis. Columns present unweighted summary statistics across the sample of automating and nonautomating banks. For nonautomating banks, the rows "Before Automation" and "After Automation" both show full sample statistics.

	Automating Banks N = 20			Nonautomating Banks N = 3,541		
	Mean	P50	SD	Mean	P50	SD
Number of Loans	3,748	1,250	6,062	511	187	1,634
Assets (Millions \$)	8,673	1,233	15,070	1,343	369	4,753
Loan Amount (\$)	131,297	129,011	61,355	77,820	59,044	73,927
Share Loans After Automation	29.9%	17.6%	31.3%			
Asian-Owned Share Before Automation	5.4%	5.1%	3.5%	3.5%	1.8%	5.8%
Asian-Owned Share After Automation	7.3%	5.4%	10.4%	3.5%	1.8%	5.8%
Black-Owned Share Before Automation	4.4%	3.2%	5.0%	2.7%	1.0%	4.8%
Black-Owned Share After Automation	12.0%	11.4%	11.6%	2.7%	1.0%	4.8%
Hispanic-Owned Share Before Automation	5.3%	3.6%	6.8%	3.0%	1.1%	7.4%
Hispanic-Owned Share After Automation	6.4%	5.5%	5.9%	3.0%	1.1%	7.4%
White-Owned Share Before Automation	85.0%	87.7%	13.4%	90.8%	94.3%	11.5%
White-Owned Share After Automation	74.4%	72.6%	21.5%	90.8%	94.3%	11.5%

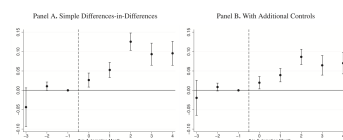


Figure 2. Share of loans to Black-owned businesses before and after small-bank automation. This figure reports dynamic difference-in-differences estimates at the monthly level and requires 11. Panel A follows the standard event-study approach to the automation event. Panel A includes fixed effects for the bank and week of loan approval. Panel B adds the vector of controls included in Table 1, column (2). We do not include more than three months before automation because the automation dates in late Spring 2020 and in late Fall 2020 following a large gap in the PPP program mean that we observe essentially no loans made four months prior to automation dates. Standard errors are clustered by zip code. The gray bars represent 95% confidence intervals.

5.3 Table III and Figure 2: bank automation event study

Read:

- Table III shows that automating banks’ Black-owned loan share rose from 4.4% before automation to 12.0% after automation.
- Figure 2 shows no differential pretrend before automation.
- After automation, the event-study coefficients turn positive and remain positive.
- Table V translates this into a controlled diff-in-diff estimate of 6.0 percentage points without controls and 4.3 percentage points with the full control set.

Economic message. This is the causal heart of the paper. The Black-owned lending share rises sharply when a bank automates, even after controlling for time and borrower composition.

5.4 Table IV and Figure 3: how much controls explain, and where the remaining gap sits

Read:

- In Table IV, Black-owned firms are 39.7 percentage points more likely than others to get a fintech PPP loan unconditionally.
- With approval-week, loan-size, location, industry, business-type, and employer controls, the gap still equals 12.1 percentage points.

Table IV
Business Owner Race and PPP Lender Type

This table reports estimates of equation (2). The dependent variable in Panel A is an indicator for whether the originating lender is a fintech firm. Panel B repeats the specifications in columns (1) and (7) of Panel A, using as dependent variables indicators for whether the originating lender is a top 4 bank (columns (1) and (2)), a large bank (columns (3) and (4)), and a small/medium-sized bank (columns (5) and (6)). Panel C is estimated on the sample of PPP loans matched to bank branch-level software spending data. The dependent variable, the log of “Bank Branch Software Spend,” is a measure of automation for the bank branches of the PPP lender within the PPP borrower’s zip code (or county, if there is no zip code match). Control variables generally pertain to the borrower firm and their particular PPP loan. Loan Amount FE are 100 indicator variables for each percentile of the loan size distribution. Zip Code and Census Tract FE are indicators for each zip code and census tract. Approval Week FE are indicators for the week in which the PPP loan was approved by the SBA. Industry FE are 104 indicators for NAICS three-digit classifications that appear in the data. Business type FE are seven indicators for the firm’s business type. Employer status is an indicator for whether the firm has at least one employee. Additional controls in Panel C are indicators for each percentile of lender assets (“Bank Size”) and branch-level revenue (“Branch Size”). Standard errors are clustered by borrower zip code. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Fintech PPP loan							
Dependent Variable:	1(Fintech)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Black-Owned)	0.397*** (0.005)	0.260*** (0.003)	0.232*** (0.003)	0.207*** (0.002)	0.219*** (0.003)	0.224*** (0.003)	0.121*** (0.002)
Approval Week FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	No	No	Yes	No	No	No	Yes
Zip Code FE	No	No	No	Yes	No	No	Yes
Industry FE	No	No	No	No	Yes	No	Yes
Business Type FE	No	No	No	No	No	Yes	Yes
Employer Status FE	No	No	No	No	No	Yes	Yes
Dep Var Mean	0.174	0.174	0.174	0.174	0.174	0.174	0.174
R ²	0.086	0.227	0.240	0.276	0.265	0.272	0.356
Observations	4,183,623	4,183,623	4,183,623	4,183,623	4,183,623	4,183,623	4,183,623

Panel B: Bank PPP loan							
Dependent Variable:	1(Top 4 Bank)		1(Large Bank)		1(Small/Med Bank)		
	(1)	(2)	(3)	(4)	(5)	(6)	
1(Black-Owned)	-0.049*** (0.002)	-0.008*** (0.001)	-0.039*** (0.001)	-0.025*** (0.001)	-0.311*** (0.004)	-0.082*** (0.001)	
Approval Week FE	No	Yes	No	Yes	No	Yes	
Loan Amount FE	No	Yes	No	Yes	No	Yes	
Zip Code FE	No	Yes	No	Yes	No	Yes	
Industry FE	No	Yes	No	Yes	No	Yes	
Business Type FE	No	Yes	No	Yes	No	Yes	
Employer Status FE	No	Yes	No	Yes	No	Yes	
Dep Var Mean	0.159	0.159	0.094	0.094	0.498	0.498	
R ²	0.001	0.317	0.001	0.131	0.030	0.396	
Observations	4,183,623	4,183,623	4,183,623	4,183,623	4,183,623	4,183,623	

(Continued)

Table IV—Continued

Panel C: Branch software spending and PPP loans to Black-owned businesses						
Dependent Variable:	Log(Branch Software Spending)					
	(1)	(2)	(3)	(4)	(5)	(6)
1(Black-owned)	0.071*** (0.006)	0.032*** (0.004)	0.025*** (0.004)	0.025*** (0.004)	0.015*** (0.003)	0.006** (0.003)
Approval Week FE	No	Yes	Yes	Yes	Yes	Yes
Borrower County FE	No	Yes	Yes	Yes	No	No
Loan Amount FE	No	Yes	Yes	Yes	Yes	Yes
Industry FE	No	Yes	Yes	Yes	Yes	Yes
Business Type FE	No	Yes	Yes	Yes	Yes	Yes
Employer Status FE	No	Yes	Yes	Yes	Yes	Yes
Branch Size FE	No	No	Yes	Yes	Yes	Yes
Bank Size FE	No	No	No	Yes	No	Yes
Bank FE	No	No	No	No	Yes	No
Borrower Zip FE	No	No	No	No	No	Yes
Dep Var Mean	10.799	10.799	10.799	10.799	10.799	10.799
Observations	2,928,221	2,928,221	2,928,221	2,928,221	2,928,221	2,928,221

- The same table shows that, after controls, the disparity is basically gone at top-4 banks but remains large at small and medium banks.
- Figure 3 shows the conditional lender-type pattern graphically across Black-, White-, Hispanic-, and Asian-owned firms.

Economic message. Observable size and geography explain a lot, but the residual gap is specifically a *small-bank versus fintech* pattern, not a generic Black-owned-firm pattern across all banks.

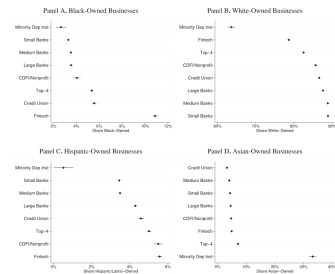


Figure 3. Conditional share of PPP loans to each race by institution type. This figure shows shares of PPP loans made to Black-owned businesses by originating lender type. Each graph presents β coefficients from variations of the following regression: Black-owned, π (Lender Type) $\times X_i + \epsilon_i$, where X_i is a vector of fixed effects for borrower zip code, loan amount percentile (in 100 bins), approval week, three-digit NAICS industry, business type, and employer status. Standard errors are clustered by zip code. The mean of the omitted category, small banks, is added back to each panel. In each panel, we change the dependent variable to be an indicator for whether a borrower is a Black-owned (Panel A), White-owned (Panel B), Hispanic-owned (Panel C), or Asian-owned (Panel D) business, multiplied by 100 to facilitate interpretation of coefficients.

Table V
Effect of Automation during PPP on Lending to Black-Owned Small Businesses
This table reports estimates of equation (3), estimated on the sample of PPP loans exclusively small and medium banks. Columns (1) and (2) of Panel A show the effect of automation on the probability that a loan is extended to a Black-owned business. Columns (3) to (6) measure effects on lending to Hispanic-, Asian-, and White-Owned businesses, respectively, using the fully controlled model from Panel A, column (2). All racial/ethnic measures are standardized to their respective levels of geography, weighted by the number of PPP loans. Controls are as described in Table IV. Standard errors are clustered by zip code. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable:	1(Black-Owned)		1(Owned by)			
	(1)	(2)	Hispanic (3)	Asian (4)	White (5)	Other (6)
1(After Automation)	0.040*** (0.007)	0.025*** (0.007)	0.000*** (0.007)	0.000*** (0.007)	0.000*** (0.007)	0.000*** (0.007)
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Approval Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep Var Mean	0.087	0.087	0.087	0.087	0.087	0.087
Observations	2,090,197	2,090,197	2,090,197	2,090,197	2,090,197	2,090,197

(Continued)

Table V—Continued

Dependent Variable:	1(Black-Owned)					
	(1)	(2)	(3)	(4)	(5)	(6)
1(After Automation)	0.041*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)
1(After Automation) + Racial Autocor.	0.041*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)	0.025*** (0.007)
Racial Autocor. Measure	1AT (logit)	1AT (logit)	1AT (logit)	1AT (logit)	1AT (logit)	1AT (logit)
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Approval Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep Var Mean	0.087	0.087	0.087	0.087	0.087	0.087
Observations	2,090,197	2,090,197	2,090,197	2,090,197	2,090,197	2,090,197

5.5 Table V and Table VI: automation, racial animus, and the discrimination channel

Read:

- Table V, Panel A shows that automation increases the Black-owned loan share by 4.3 percentage points with full controls, while also slightly increasing Hispanic- and Asian-owned shares and lowering the White-owned share.
- Table V, Panel B shows that the automation effect is stronger in places with more anti-Black animus.
- Table VI shows that Black-owned firms are more likely to borrow from fintechs in high-animus places and less likely to borrow from non-top-4 banks there.
- For the implicit-bias IAT measure, a one-standard-deviation increase in animus implies about 1.3 percentage points more fintech borrowing and about 2.2 percentage points less non-top-4-bank borrowing for Black-owned firms.

Table VI
Black Business Ownership and Lender Identity: The Effect of Racial Animus

This table reports estimates of a modified equation (2), focusing on the interaction between the indicator for Black-owned business and a standardized measure of racial animus in the borrower location. The dependent variable differs across the three panels. In Panel A it is an indicator for a fintech PPP loan, in Panel B it is an indicator for a non-top 4 bank PPP loan, and in Panel C it is an indicator for a top 4 bank PPP loan. In each panel, column (1) repeats the specification in Table IV, Panel A, column (7). The racial animus measures are as follows: columns (2) to (3) use the implicit and explicit score from the Implicit Association Test (IAT) aggregated to the county level; column (4) uses the number of racially charged searches in a designated media market (DMA); column (5) uses responses to the question on favorability toward Black people in the Nationscape survey aggregated to the congressional district level; and columns (6) to (7) use the dissimilarity and isolation index at the metropolitan statistical area (MSA) level. All racial animus measures are standardized at their respective levels of geography, weighted by the number of PPP loans. Controls are as described in Table IV. Standard errors are clustered by zip code. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Fintech PPP loans as dependent variable							
Dependent Variable:	1(Fintech)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Black-Owned)	0.121*** (0.002)	0.120*** (0.001)	0.121*** (0.001)	0.124*** (0.001)	0.121*** (0.002)	0.121*** (0.001)	0.107*** (0.001)
1(Black-Owned) × Racial Animus		0.013*** (0.002)	0.011*** (0.002)	0.004** (0.002)	0.014*** (0.002)	0.016*** (0.002)	0.028*** (0.002)
Dep Var Mean	0.174		0.174		0.174		0.174

Panel B: Non-Top 4 bank PPP loan as dependent variable							
Dependent Variable:	1(Non-Top 4 Bank)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Black-Owned)	-0.107*** (0.002)	-0.105*** (0.001)	-0.107*** (0.001)	-0.108*** (0.001)	-0.107*** (0.001)	-0.104*** (0.002)	-0.094*** (0.001)
1(Black-Owned) × Racial Animus		-0.022*** (0.002)	-0.014*** (0.002)	-0.013*** (0.002)	-0.012*** (0.002)	-0.012*** (0.002)	-0.028*** (0.002)
Dep Var Mean	0.592		0.592		0.592		0.592

(Continued)

Table VI—Continued

Panel C: Top 4 bank PPP loans as dependent variable

Dependent Variable:	1(Top 4)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Black-Owned)	-0.008*** (0.001)	-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)
1(Black-Owned) × Racial Animus		0.006*** (0.001)	0.004*** (0.001)	0.010*** (0.001)	-0.001 (0.001)	-0.004*** (0.001)	0.004*** (0.001)
Racial Animus Measure		IAT (Implicit)	IAT (Explicit)	Stephens-Davidowitz	Nationscape	Segregation (Dissimilar-ity)	Segregation (Isolation)
Approval Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep Var Mean	0.159	0.159	0.159	0.159	0.159	0.159	0.159
Observations	4,185,623	4,185,623	4,185,623	4,185,623	4,185,623	4,185,623	4,185,623

Economic message. The paper’s most persuasive evidence on discrimination is not the average gap by itself, but the fact that the gap lines up systematically with local racial animus and shrinks when human discretion is removed.

5.6 Table VII: Lendio application data

Read:

Table VII
Lending to Black-Owned Firms Conditional on Applications

This table reports estimates of a modified version of equation (2) within the sample of PPP loan application to Lendio. Columns (1) and (2) include the full sample, columns (3) and (4) restrict attention to the subsample of applications that Lendio quasi-randomly forwarded only to fintech lenders, columns (5) and (6) restrict attention to the subsample of loans that were forwarded only to conventional lenders, and columns (7) and (8) restrict attention to the subsample of loans that were forwarded only to small banks. In each sample, we consider two dependent variables: an indicator for whether the application eventually resulted in a PPP loan from one of the lenders who Lendio forwarded the application to (columns (1), (3), (5), and (7)), and an indicator variable for whether the loan applicant received no PPP loan at all (through a Lendio lender or an alternative lender). We include fixed effects for each combination of lenders who the application was routed to. Other controls are as described in Table IV, with the exception that we include application week fixed effects instead of origination week fixed effects. Standard errors are clustered by zip code. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Sent to Lenders:	Any		Only Fintech		Only Conventional		Only Small Banks	
	1(PPP Loan from Lendio Lender)	1(No PPP Loan)	1(PPP Loan from Lendio Lender)	1(No PPP Loan)	1(PPP Loan from Lendio Lender)	1(No PPP Loan)	1(PPP Loan from Lendio Lender)	1(No PPP Loan)
Dependent Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1(Black-owned)	-0.011*** (0.003)	0.021*** (0.002)	-0.000 (0.004)	0.006** (0.003)	-0.038*** (0.005)	0.058*** (0.005)	-0.056*** (0.010)	0.068*** (0.009)
Application Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sent Lenders FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dep Var Mean	0.250	0.396	0.239	0.399	0.316	0.364	0.382	0.375
Observations	278,404	278,404	153,434	153,434	83,328	83,328	36,276	36,276

- Among all Lendio applications, Black-owned firms are less likely to get a PPP loan from the routed lender and more likely to get no PPP loan at all.
- When Lendio routes applications only to fintech lenders, the racial approval gap is essentially zero.
- When applications are routed only to conventional lenders, Black-owned firms are 3.9 percentage points less likely to get the loan from that lender and 5.8 percentage points more likely to get no PPP loan at all.
- When routed only to small banks, the disparities are even larger.

Economic message. This table is the main evidence that lender-side processing and approval behavior matters, not just where firms chose to apply.

5.7 Table VIII and Table IX: relationship banking does not rescue the bank story

Table VIII
Black Business Ownership and PPP Lender Type with Bank and Credit Relationship Controls

This table reports estimates of a modified equation (2), focusing on the role of bank and credit relationships. The sample is restricted to bank statement-matched data. We include only information from a firm's latest statement prior to the loan approval. The dependent variable in Panel A is an indicator for whether a PPP loan is originated by a fintech lender. The dependent variables in Panel B are indicators for whether the originating lender is a top 4 bank (columns (1) and (2)), a large bank (columns (3) and (4)), or a small/medium-sized bank (columns (5) and (6)). We report coefficients on indicators for whether the borrower has previous credit relationships with fintech and nonfintech lenders. Checking Acct Bank FE are indicators for the bank where the borrower has its main business checking account, so that we compare borrowers who bank with the same institution. Monthly Net Cash Inflow FE and Monthly Cash Inflow FE are each a set of 100 percentile indicators for monthly net cash inflow and total cash inflow, respectively. Other controls are as described in Table IV. Standard errors are clustered by borrower zip code. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Fintech PPP loan				
Dependent Variable:	1(Fintech)			
	(1)	(2)	(3)	(4)
1(Black-Owned)	0.055*** (0.004)	0.055*** (0.004)	0.056*** (0.004)	0.055*** (0.004)
1(Credit from Fintech)			0.075*** (0.003)	0.078*** (0.003)
1(Credit from Conv.)			-0.012*** (0.003)	-0.011*** (0.003)
Approval Week FE	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes
Months Since Statement FE	No	Yes	Yes	Yes
Checking Acct Bank FE	No	Yes	Yes	Yes
Monthly Cash Inflow FE	No	No	No	Yes
Monthly Net Cash Inflow FE	No	No	No	Yes
Dep Var Mean	0.363	0.363	0.363	0.363
Observations	168,360	168,360	168,360	168,360

(Continued)

Table VIII—Continued

Panel B: Bank PPP loan						
Dependent Variable:	1(Top 4 Bank)		1(Large Bank)		1(Small/Med Bank)	
	(1)	(2)	(3)	(4)	(5)	(6)
1(Black-Owned)	0.005 (0.003)	-0.000 (0.003)	-0.017*** (0.003)	-0.019*** (0.002)	-0.047*** (0.004)	-0.037*** (0.003)
1(Credit from Fintech)			-0.025*** (0.003)	-0.011*** (0.002)		-0.035*** (0.003)
1(Credit from Conv.)			0.001 (0.002)	0.005** (0.002)		0.001 (0.003)
Approval Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes	Yes	Yes
Months Since Statement FE	No	Yes	No	Yes	No	Yes
Checking Acct Bank FE	No	Yes	No	Yes	No	Yes
Monthly Cash Inflow FE	No	Yes	No	Yes	No	Yes
Monthly Net Cash Inflow FE	No	Yes	No	Yes	No	Yes
Bank Statement Sample	Latest	Latest	Latest	Latest	Latest	Latest
Dep Var Mean	0.177	0.177	0.090	0.090	0.304	0.304
Observations	168,360	168,360	168,360	168,360	168,360	168,360

Read:

- In the Oculous bank-statement sample, Black ownership predicts a 5.5 percentage-point higher probability of a fintech PPP loan even in the matched sample.
- Adding checking-account-bank fixed effects leaves that estimate nearly unchanged.
- Prior credit relationships matter for lender identity in the expected directions, but they do not eliminate the racial disparity.
- The disparity is especially strong among firms that borrow from a lender other than their checking-account bank.

Table IX

Black Business Ownership and PPP Lender Type by Checking Account Bank Type

This table reports estimates of a modified equation (2), focusing on various samples of firms with different checking account bank and credit relationships. Panels A, B, and C limit the sample to PPP borrowers with checking accounts at top 4 banks, non-top 4 large banks, and small/medium banks, respectively. Panel D limits the sample to PPP borrowers who have previous credit relationships with fintech lenders. Across all panels, the dependent variable in column (1) is an indicator for whether a PPP loan is originated by the borrower's checking account bank. The dependent variables in columns (2)–(5) are indicators for whether a PPP loan is originated by a fintech lender, top 4 bank, non-top 4 large bank, and small/medium bank, respectively. Controls are as described in Table VIII. Standard errors are clustered by borrower zip code. ***, **, *, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dep Var:	1(Lender is Bank) (1)	1(Fintech) (2)	Banks:		
			1(Top 4) (3)	1(Large) (4)	1(Small/Medium) (5)
Panel A: Sample of borrowers with checking accounts at top 4 banks					
1(Black-Owned)	−0.006 (0.005)	0.051*** (0.006)	−0.007 (0.005)	−0.010*** (0.003)	−0.034*** (0.004)
Observations	80,560	80,560	80,560	80,560	80,560
Dep Var Mean	0.245	0.492	0.310	0.048	0.188
Panel B: Sample of borrowers with checking accounts at non-top 4 large banks					
1(Black-Owned)	−0.024** (0.010)	0.044*** (0.011)	0.009 (0.006)	−0.033*** (0.010)	−0.041*** (0.009)
Observations	26,539	26,539	26,539	26,539	26,539
Dep Var Mean	0.381	0.364	0.062	0.309	0.216

(Continued)

Table IX—Continued

Panel C: Sample of borrowers with checking accounts at small/medium-sized banks					
1(Black-Owned)	−0.015 (0.010)	0.056*** (0.010)	−0.001 (0.005)	−0.008 (0.005)	−0.044*** (0.010)
Observations	46,352	46,352	46,352	46,352	46,352
Dep Var Mean	0.355	0.256	0.051	0.053	0.591
Panel D: Sample of borrowers with fintech credit relationship					
1(Black-Owned)	0.019 (0.014)	0.032** (0.014)	0.016 (0.012)	−0.009 (0.009)	−0.040*** (0.011)
Observations	23,890	23,890	23,890	23,890	23,890
Dep Var Mean	0.206	0.339	0.194	0.065	0.311
Approval Week FE	Yes	Yes	Yes	Yes	Yes
Loan Amount FE	Yes	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes
Employer Status FE	Yes	Yes	Yes	Yes	Yes
Months Since Statement FE	Yes	Yes	Yes	Yes	Yes
Checking Acct Bank FE	Yes	Yes	Yes	Yes	Yes
Monthly Cash Inflow FE	Yes	Yes	Yes	Yes	Yes
Monthly Net Cash Inflow FE	Yes	Yes	Yes	Yes	Yes
Credit Rel. Controls	Yes	Yes	Yes	Yes	Yes

Economic message. Conventional banks did favor their own customers, but that does not explain why Black-owned firms ended up at fintechs. The bigger problem was getting served as a *new* customer under processing pressure.

6 Short summary

- The paper studies racial disparities in PPP lending through the lens of lender automation.
- Fintechs and highly automated banks made a much larger share of PPP loans to Black-owned firms than small conventional banks.
- This reflects three channels: lower fixed cost per loan, broader geographic reach, and less human discretion.
- Rich controls cut the Black-owned fintech tilt substantially but do not eliminate it.
- Within-bank evidence shows that automation causally raises Black-owned lending shares.
- The remaining cross-lender gap is strongest in high-animus locations and shows up in lender approvals, not just in borrower applications.
- Preexisting banking relationships, contemporaneous firm performance, and fraud do not explain the main results.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that process automation reduced racial disparities in access to PPP credit. It does so most cleanly in the within-bank automation design, where Black-owned lending shares rise sharply once banks automate. The broader cross-lender evidence shows that the least automated lenders—especially small banks—served Black-owned firms at much lower rates than fintechs, and that this difference cannot be fully explained by firm size, geography, application choices, prior bank relationships, or firm performance.

7.2 Economic intuition

Automation changes lending outcomes because PPP lending was a high-volume, capacity-constrained processing problem. When applications flood in and fees are low on small loans, lenders with manual systems will ration attention in ways that favor larger, easier, and possibly more socially familiar borrowers. Automated systems are better at cheaply processing small loans, reaching firms without branch relationships, and removing human discretion from bottlenecks such as intake, verification, and prioritization.

7.3 Takeaway

The paper's deepest message is that technology can change the distribution of credit not only by better predicting risk, but also by changing *who gets processed in the first place*. In that sense, automation is not just a productivity tool. It is also an organizational design choice with real consequences for equity in credit access.